

# Network Science

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## Lecture 04: Communities and Graph Partitioning

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From single ties and nodes to groups, roles, and community structure.

# Communities and Graph Partitioning

Today we move from single ties and nodes to groups, roles, and community structure.

## Focus

- communities, closure, and structural holes
- centrality measures as role concepts
- community detection vs. graph partitioning
- hierarchical clustering and Girvan-Newman

# Module 1

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## Roles and Social Capital

**Guiding Question:** What kinds of structurally different positions can people occupy inside the same network?

# Formal Distinction

Not all important positions are the same.

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- : a node connects otherwise separated groups
- : a gap between groups that brokerage can span

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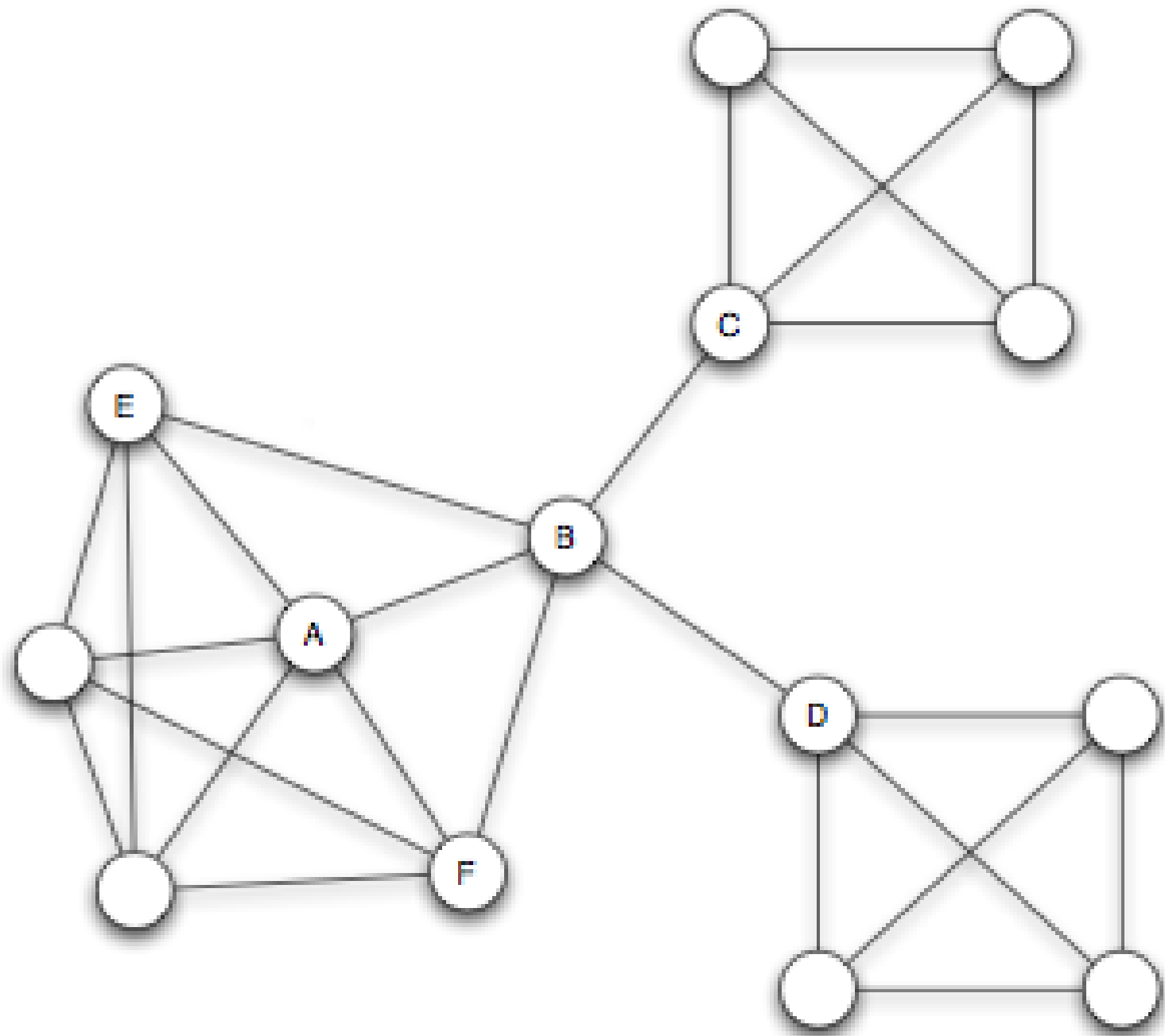
Not all important positions are the same.

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# Structural Roles







Source: Burt 1992

# Interpretation

# Interpretation

- embedded nodes sit inside dense trusted neighborhoods

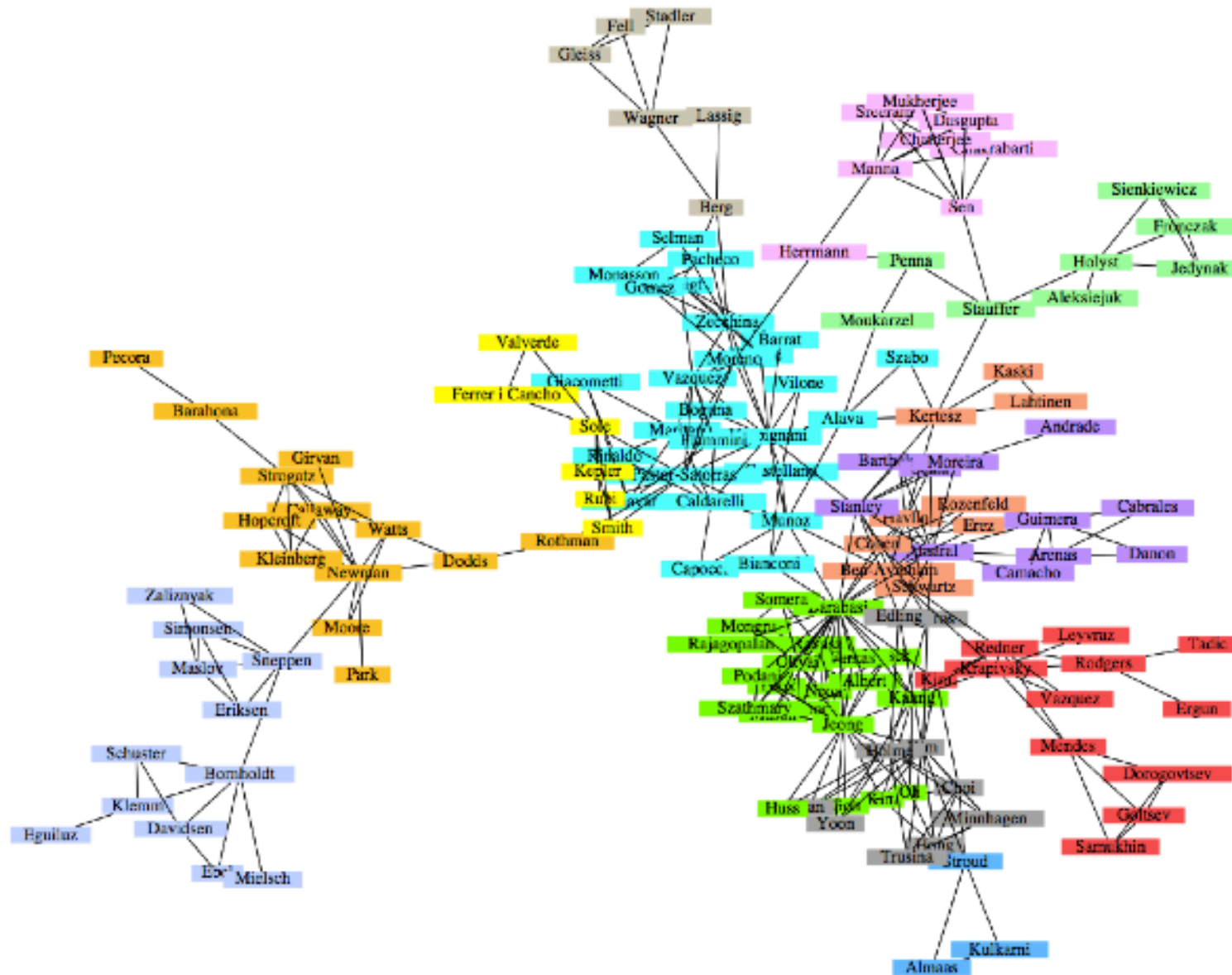
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- embedded nodes sit inside dense trusted neighborhoods
- structural holes connect otherwise weakly linked groups
- highly connected nodes may have power through scale, not only brokerage

# Communities



Source: Easley & Kleinberg 2010

Community structure means the network contains groups of tightly interconnected nodes.

# Mini-Exercise 4.A: Roles in a Karate Club

In Zachary's karate club, two factions form around the instructor (Mr. Hi) and the administrator (Officer).

1. Which structural role best describes Mr. Hi: embedded, broker, or hub?
2. Are there any nodes that connect the two factions? What role do they play?



# Mini-Exercise 4.A: Solution

Hub and embedded — high degree within his faction, central to a tightly connected subgroup.

A small number of members maintain ties across the factions. They occupy brokerage positions and span the structural hole between the two groups. When the club split, these were the nodes whose loyalty was hardest to predict.

# Mini-Exercise 4.A: Solution

**1. Mr. Hi:** Hub and embedded — high degree within his faction, central to a tightly connected subgroup.

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# Mini-Exercise 4.A: Solution

1. **Mr. Hi:** Hub and embedded — high degree within his faction, central to a tightly connected subgroup.
2. **Bridging nodes:** A small number of members maintain ties across the factions. They occupy brokerage positions and span the structural hole between the two groups. When the club split, these were the nodes whose loyalty was hardest to predict.

# Takeaway

"Importance" is not one thing. Brokerage, embeddedness, and degree capture different social advantages.

# Module 2

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## Centrality

**Guiding Question:** How should we measure which nodes are important?

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importance  $\neq$  degree alone

# Formal View

Different centrality measures formalize different intuitions about importance.

- : how many links a node has
- : how near a node is to all others
- : how often a node lies on shortest paths
- : how strongly a node is connected to other important nodes

# Formal View

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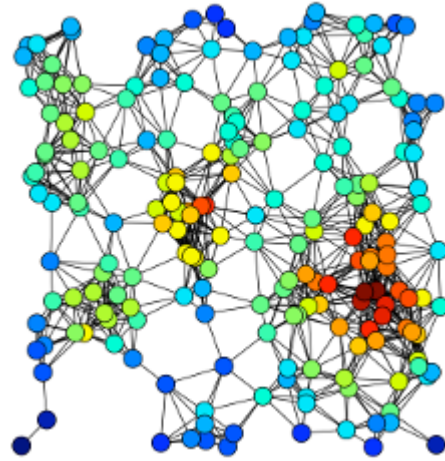
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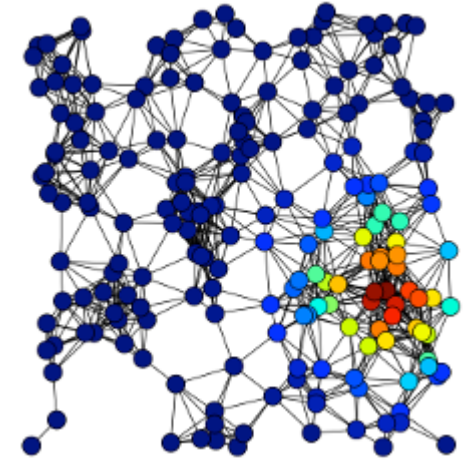
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# Degree-Based Views

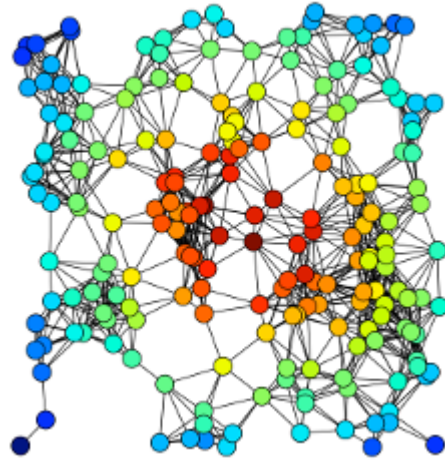


*Source: Easley & Kleinberg 2010*

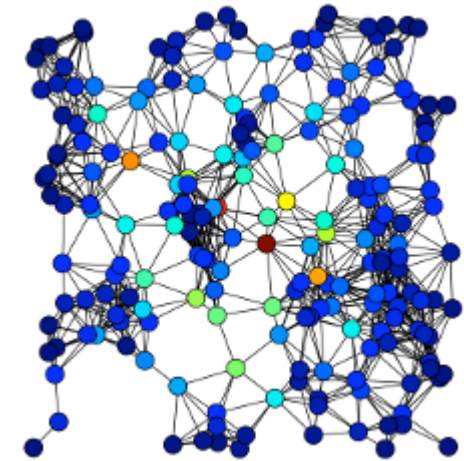


*Source: Easley & Kleinberg 2010*

# Distance and Brokerage



*Source: Easley & Kleinberg 2010*



*Source: Easley & Kleinberg 2010*

# Interpretation

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# Interpretation

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- who has many links?
- who is linked to important others?
- who is close to everyone?
- who controls paths between groups?

# Mini-Exercise 4.B: Choose the Right Centrality

For each scenario, decide which centrality measure is most appropriate and justify briefly.

1. You want to identify the most influential gossip in a small village.
2. You want to find emergency-response hubs that should reach all districts within minimum time.
3. You want to detect which router, if attacked, would slow internet traffic between continents most.
4. You want to find the most prestigious researcher in a citation network.

# Mini-Exercise 4.B: Solution

1. — gossip is local, direct contact matters.
2. — minimizing average distance to all districts.
3. — the router's role is to lie on paths between continents.
4. — prestige flows from being cited by others who are themselves prestigious.

# Mini-Exercise 4.B: Solution

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2. **Closeness centrality** — minimizing average distance to all districts.
3. **Betweenness centrality** — the router's role is to lie on paths between continents.
4. **Eigenvector centrality / PageRank** — prestige flows from being cited by others who are themselves prestigious.



# Takeaway

Centrality is always tied to a theory of influence, access, or control. Choosing a measure is choosing a theory.

# Module 3

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## Communities

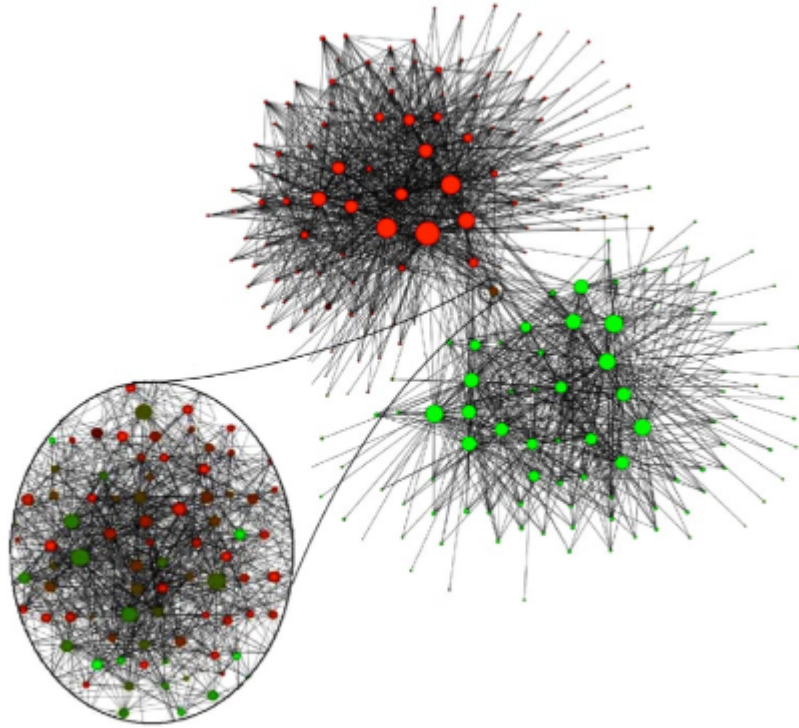
**Guiding Question:** What should count as a community in a graph?

# Formal Intuition

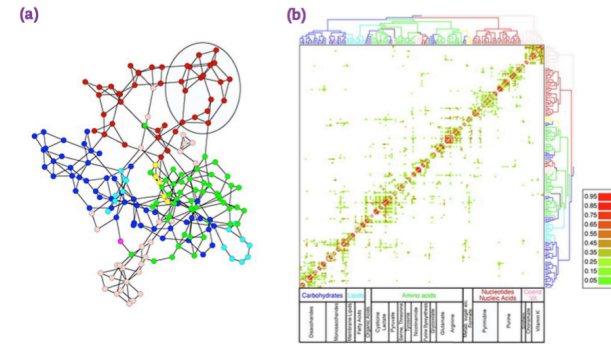
A community is a subset of nodes with:

- relatively many edges inside the subset
- relatively few edges leaving the subset
- a density that is meaningfully higher than the surrounding graph

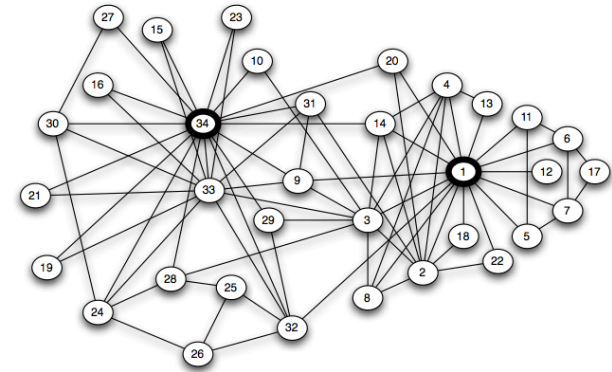
# Example Cases



Source: Easley & Kleinberg 2010



Source: Easley & Kleinberg 2010



Source: Zachary 1977 — modified; Original data: Zachary 1977; visualization redrawn

# Working Definition

A community is a locally dense connected subgraph whose internal connectivity is stronger than its connectivity to the outside.

# Interpretation

Graph partitioning and community detection are related but not identical:

- **partitioning** usually fixes the number or size of groups
- **community detection** usually lets the structure determine the grouping
- the choice of objective function changes what counts as a good community

# Mini-Exercise 4.C: When Partitioning Fails

A graph has two large dense clusters connected by a single edge, plus one small triangle attached to one of the clusters via a single edge.

1. How many "natural" communities does this graph have?
2. If you ask a balanced 2-partition algorithm to split it into two equal halves, what might go wrong?

# Mini-Exercise 4.C: Solution

Three — the two dense clusters and the small triangle.

A balanced 2-partition has to split nodes into equal-sized groups. It will either cut through one of the dense clusters or merge the triangle with the wrong cluster — neither matches the natural structure. The constraint distorts the answer.



# Mini-Exercise 4.C: Solution

**1. Natural communities:** Three — the two dense clusters and the small triangle.

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# Mini-Exercise 4.C: Solution

1. **Natural communities:** Three — the two dense clusters and the small triangle.
2. **What goes wrong:** A balanced 2-partition has to split nodes into equal-sized groups. It will either cut through one of the dense clusters or merge the triangle with the wrong cluster — neither matches the natural structure. The constraint distorts the answer.

# Takeaway

Communities are not visual artifacts. They are claims about density, separation, and interpretability — and they depend on the objective function used to find them.

# Module 4

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## Hierarchical Clustering

**Guiding Question:** How can we discover communities without fixing them in advance?

# Core Idea



# Core Idea

merge or split groups according to similarity

# Formal View

Hierarchical clustering builds a tree of nested groups.

At each step it uses a similarity or distance rule to decide which clusters to merge or split.

# Two Strategies

- : start with single nodes and merge
- : start with the whole graph and split



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- **agglomerative**: start with single nodes and merge
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# Two Strategies

- **agglomerative:** start with single nodes and merge
- **divisive:** start with the whole graph and split
- both produce a hierarchy rather than a single flat answer

# Interpretation

The key modeling choice is the similarity or distance function:

- first-order similarity uses direct links
- second-order similarity uses neighborhood overlap
- linkage rules define how cluster-to-cluster distances are updated

# Mini-Exercise 4.D: Choosing a Cutoff

A dendrogram for a friendship network shows clear merges at three levels: 5 groups at one level, 3 groups at another, 2 groups at the highest level.

1. Which level captures the "right" community structure?
2. What information would you need to decide?

# Mini-Exercise 4.D: Solution

There is no single correct level — each is "right" for a different question.

- the of the grouping (assignment, recommendation, diagnosis)

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- external validation: known group labels, modularity scores, or task performance

# Mini-Exercise 4.D: Solution

1. There is no single correct level — each is "right" for a different question.

2. Information needed:

- the *purpose* of the grouping (assignment, recommendation, diagnosis)
- external validation: known group labels, modularity scores, or task performance
- domain knowledge about plausible group sizes

# Takeaway

Hierarchical clustering is useful when the data may contain meaningful structure at several resolutions. The cutoff is a modeling decision, not an algorithmic output.

# Module 5

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## Girvan-Newman

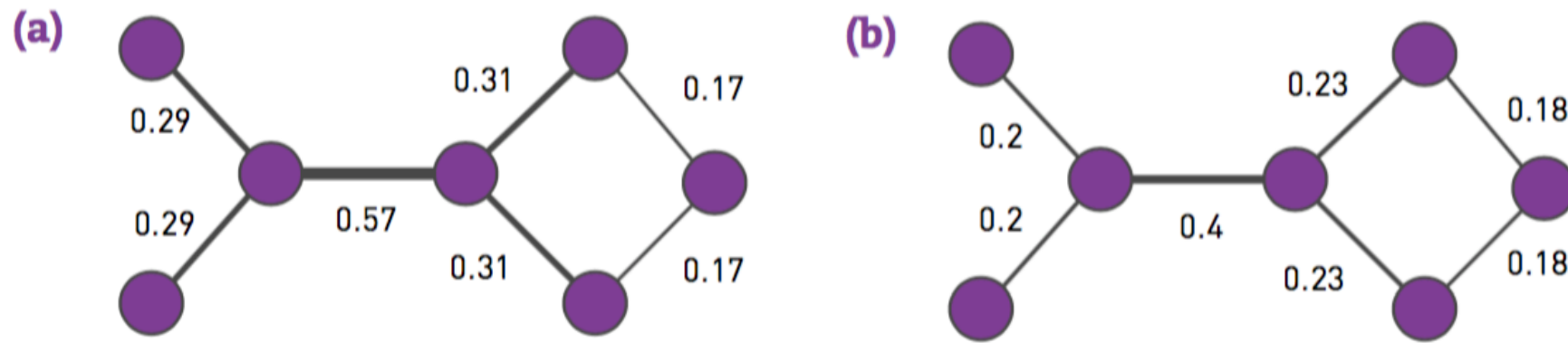
**Guiding Question:** Can communities be found by removing the edges that carry the most inter-group traffic?

# Formal Rule

Edge betweenness counts how many shortest paths pass through an edge.

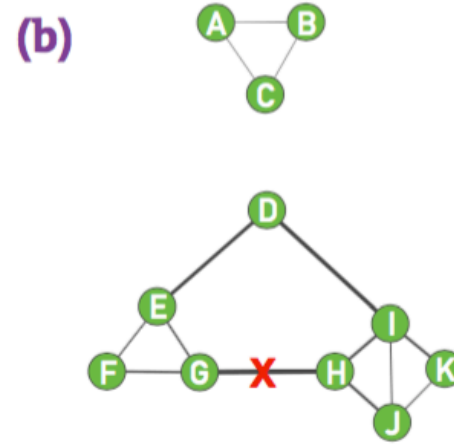
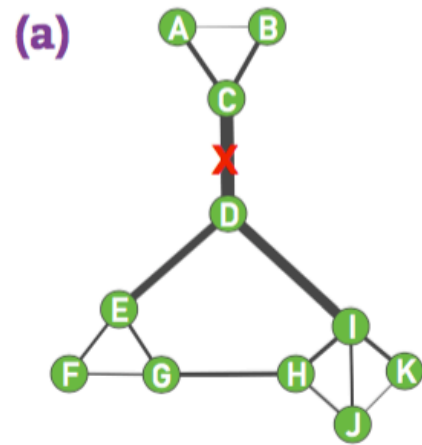
Girvan-Newman repeatedly removes the edge with the highest edge betweenness.

# Core Result

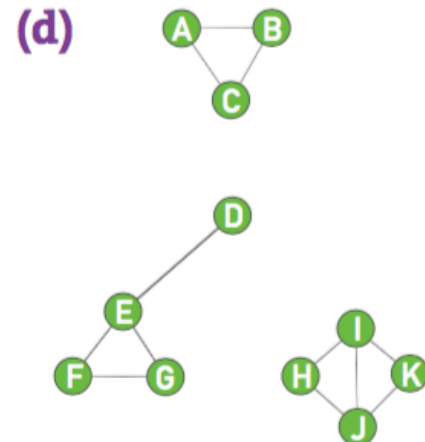
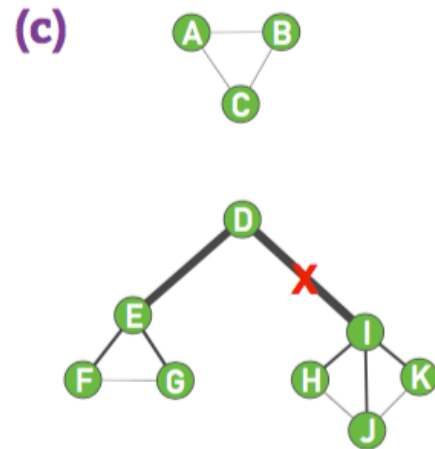
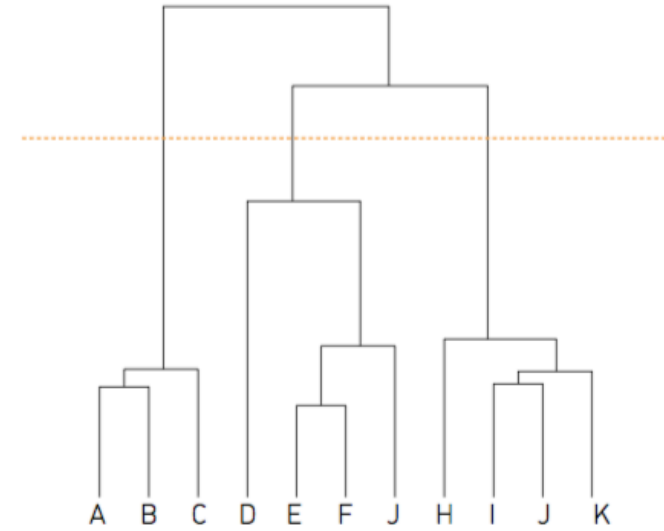


Source: Girvan & Newman 2002

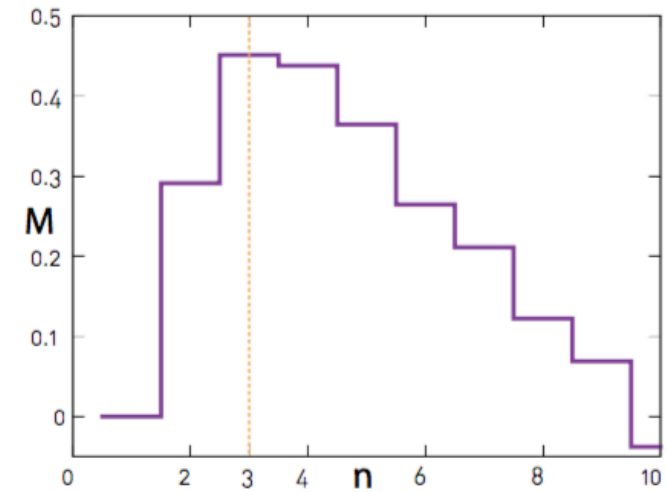
# Example



(e)



(f)



Source: Girvan & Newman 2002



# Interpretation

Edges that sit between communities tend to be crossed by many shortest paths, so removing them can reveal hidden group structure.

# Mini-Exercise 4.E: Girvan-Newman by Hand

In a small graph, the edge with the highest betweenness has been removed. The graph is now split into two components of roughly equal size, plus one isolated triangle.

1. Should you stop here or continue removing edges?
2. What would tell you the algorithm has reached a "good" stopping point?

# Mini-Exercise 4.E: Solution

1. Continue — the algorithm produces a sequence of nested partitions. Each removal step gives a new candidate clustering.
2. **Stopping criterion:** Use  $Q$  — measure how much denser the within-community edges are than expected under a random null model. Pick the level that maximizes  $Q$ .

# Mini-Exercise 4.E: Solution

1. Continue — the algorithm produces a *hierarchy* of nested partitions. Each removal step gives a new candidate clustering.
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# Mini-Exercise 4.E: Solution

1. Continue — the algorithm produces a *hierarchy* of nested partitions. Each removal step gives a new candidate clustering.
2. **Stopping criterion:** Use **modularity  $Q$**  — measure how much denser the within-community edges are than expected under a random null model. Pick the level that maximizes  $Q$ .

# Takeaway

Girvan-Newman is conceptually elegant and didactically useful, but computationally expensive for large graphs. Modern methods (Louvain, Leiden) optimize modularity directly without recomputing edge betweenness at every step.

# Wrap-Up

- centrality asks which nodes matter and why
- communities ask where the graph is dense and where it is sparse
- different methods optimize different notions of structure

## Looking Ahead: Lecture 05

If links reflect both choice and context, how do we model the social settings in which networks form?  
When does similarity cause friendship — and when does friendship cause similarity?

## Image Credits & References

Burt, Ronald S. (1992). Structural Holes: The Social Structure of Competition. *Harvard University Press*.

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